

## WT EXP -1

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**Aim:** Comparison of different Wireless Technologies based on following parameters:

**Theory:**

**1. Bluetooth:**

- Definition: A short-range wireless technology designed for exchanging data between devices over short distances, commonly used in personal area networks (PANs) for connecting devices like phones, headphones, and keyboards.
- Features: Short-range wireless communication, low power consumption, simple pairing process.
- Applications: Wireless audio devices (headphones, speakers), fitness trackers, keyboard and mouse connections, IoT devices.

**2. Wi-Fi:**

- Definition: A wireless networking technology that allows devices to connect to the internet or local area networks (LANs) using radio waves, providing fast data transfer over medium-range distances, typically within homes, offices, or public spaces.
- Features: High-speed internet access, wide coverage area, supports large data throughput.
- Applications: Internet access in homes, offices, and public spaces, streaming, gaming, file sharing.

**3. Zigbee:**

- Definition: A low-power wireless communication standard designed for IoT (Internet of Things) applications, enabling simple, low-data-rate, and long battery-life communication for devices like sensors, smart home devices, and automation systems.
- Features: Low power consumption, low data rates, scalable network with multiple devices.
- Applications: Smart home systems, industrial automation, remote sensing, energy management.

**4. LoRa:**

- Definition: A long-range, low-power wireless communication technology designed for IoT applications, enabling devices to send small amounts of data over vast distances, especially in remote areas with low energy consumption.
- Features: Long-range communication, low energy consumption, low data rates.
- Applications: Agriculture monitoring, smart city infrastructure, environmental sensing, asset tracking.

5. **5G:**

- Definition: The fifth generation of mobile network technology, designed to offer ultra-fast data speeds, low latency, and higher capacity, enabling innovations like autonomous vehicles, smart cities, and augmented reality.
- Features: Ultra-fast data speeds, low latency, high capacity, supports massive IoT connectivity.
- Applications: Autonomous vehicles, smart cities, augmented reality, high-speed mobile data.

6. **NFC:**

- Definition: Near Field Communication (NFC) is a short-range wireless technology that allows devices to exchange data by simply touching or bringing them close together, commonly used for contactless payments and data sharing.
- Features: Very short-range communication, fast data exchange, low power usage.
- Applications: Contactless payments, access control, data sharing between smartphones, public transport systems.

7. **Infrared:**

- Definition: A short-range communication technology using infrared light to transmit data between devices, commonly used in remote controls, personal area networks, and device pairings.
- Features: Short-range, line-of-sight communication, inexpensive, simple to implement.
- Applications: Remote controls, data transfer between devices, proximity sensing, personal area networks.

8. **Satellite Communication:**

- Definition: A communication system that uses satellites in orbit to transmit signals for data, voice, and internet services, providing coverage in remote or hard-to-reach areas globally.
- Features: Global coverage, reliable for remote areas, high signal latency.
- Applications: Television broadcasting, global internet access, emergency communication, maritime communication.

9. **Cellular Communication:**

- Definition: A communication system used by mobile phones and wireless devices, where signals are transmitted through a network of base stations, enabling voice, text, and data services over large areas.
- Features: Wide-area coverage, supports voice, text, and data, varying speeds based on generation (4G, 5G).
- Applications: Mobile voice calls, texting, mobile internet, location-based services.

#### 10. Fan-Out Technology:

- **Definition:** A technology that enables a central device to communicate with multiple devices simultaneously, often used in high-performance computing or distributed systems for tasks like load balancing or data distribution.
- **Features:** Scalable, allows simultaneous communication with multiple devices, used in distributed systems.
- **Applications:** Load balancing in data centers, cloud computing, distributed database management, high-performance computing.

#### Comparison Table:

| Technology              | Range                    | Cost                | Propagation Delay | Power Consumption | Throughput              |
|-------------------------|--------------------------|---------------------|-------------------|-------------------|-------------------------|
| Bluetooth               | Up to 100 meters         | ₹500 - ₹2,500       | Low               | Low               | Up to 3 Mbps            |
| Wfi                     | Up to 100 meters indoors | ₹1,000 - ₹5,000     | Low               | Medium            | Up to 1 Gbps            |
| Zigbee                  | Up to 100 meters         | ₹200 - ₹2,000       | Low               | Low               | Up to 250 kbps          |
| LoRa                    | Up to 15 kilometers      | ₹2,500 - ₹10,000    | High              | Very Low          | Up to 27 kbps           |
| 5G                      | Up to 10 kilometers      | ₹10,000 - ₹50,000   | Very Low          | High              | Up to 10 Gbps           |
| NFC                     | Up to 10 centimeters     | ₹200 - ₹2,000       | Very Low          | Very Low          | Up to 424 kbps          |
| Infrared                | Up to 1-2 meters         | ₹100 - ₹1,000       | Low               | Very Low          | Up to 4 Mbps            |
| Satellite Communication | Global                   | ₹50,000 - ₹2,00,000 | Very high         | High              | Up to 512 kbps          |
| Cellular Communication  | Up to 30 kilometers      | ₹3,000 - ₹15,000    | Low               | Medium            | Up to 100 Mbps (4G LTE) |
| Fan-Out Technology      | Not applicable           | Varies              | Low               | Varies            | High                    |

**Conclusion:**

Each wireless technology studied serves distinct use cases based on its range, power consumption, and data throughput. Technologies like **Bluetooth**, **Wi-Fi**, and **Zigbee** excel in short-range, low-power applications, while **5G** and **Satellite Communication** offer high-speed, long-range capabilities for more demanding needs. **LoRa** and **NFC** provide specialized solutions for IoT and contactless applications, respectively, balancing range, power, and cost for specific scenarios.